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“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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(1WN24CS232)**

**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
February to June 2026**

CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by **Rohan S Mirjankar (1WN24CS232)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab Program 1:

Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>

int main() {
    int n, adj[20][20], indegree[20] = {0};
    int i, j, count = 0;
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter adjacency matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++) {
            scanf("%d", &adj[i][j]);
            if(adj[i][j] == 1)
                indegree[j]++;
        }
    }
    printf("Topological Order: ");
    while(count < n) {
        for(i = 0; i < n; i++) {
            if(indegree[i] == 0) {
                printf("%d ", i);
                indegree[i] = -1;
                count++;
                for(j = 0; j < n; j++) {
                    if(adj[i][j] == 1)
                        indegree[j]--;
                }
            }
        }
    }
}
```

```

    }

    return 0;

}

```

Output:

The screenshot shows the Code::Blocks IDE with the following components:

- Main Editor:** Displays the C++ source code for 'Topological-sorting.c'. The code includes `<stdio.h>` and implements a topological sorting algorithm using an adjacency list and indegree array.
- Output Window:** A separate window titled 'D:\Sem 4\ADA\LAB\Topological-sorting.exe' shows the program's execution. It prompts for the number of vertices (5) and an adjacency matrix. The matrix is entered as:


```

0 1 1 0 0
0 0 0 1 0
0 0 0 1 0
0 0 0 0 1
0 0 0 0 0

```

 The output then shows the topological order: 0 1 2 3 4, the execution time (53.379 s), and a message to press any key to continue.
- Log & others:** A panel at the bottom shows build messages, including 'Build file: "no target" in "no project" (compiler: unknown)' and 'Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s))'.

Leet Code exercises on topological sorting

207. Course Schedule

Problem:

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you must take course b_i first if you want to take course a_i.

- For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

Code:

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize){
    int* indegree = (int*)calloc(numCourses, sizeof(int));
    // adjacency list
    int** adj = (int**)malloc(numCourses * sizeof(int*));
    int* adjSize = (int*)calloc(numCourses, sizeof(int));
    for (int i = 0; i < numCourses; i++) {
        adj[i] = (int*)malloc(numCourses * sizeof(int));
    }
    // build graph
    for (int i = 0; i < prerequisitesSize; i++) {
        int course = prerequisites[i][0];
        int prereq = prerequisites[i][1];
        adj[prereq][adjSize[prereq]++] = course;
        indegree[course]++;
    }
    // queue
    int* queue = (int*)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0)
```

```

        queue[rear++] = i;
    }

    int count = 0;

    while (front < rear) {
        int node = queue[front++];

        count++;

        for (int i = 0; i < adjSize[node]; i++) {
            int neighbor = adj[node][i];

            indegree[neighbor]--;

            if (indegree[neighbor] == 0)
                queue[rear++] = neighbor;
        }
    }

    return count == numCourses;
}

```

Output:

The screenshot displays the LeetCode interface for the "207. Course Schedule" problem. The problem description states: "There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you must take course b_i first if you want to take course a_i." It includes two examples: Example 1 where numCourses = 2 and prerequisites = [[1,0]] results in true, and Example 2 where numCourses = 2 and prerequisites = [[1,0],[0,1]] results in false. The constraints are listed as: 1 ≤ numCourses ≤ 2000, 0 ≤ prerequisites.length ≤ 5000, prerequisites[i].length == 2, 0 ≤ a_i, b_i < numCourses, and all pairs prerequisites[i] are unique.

The C++ code in the editor implements a topological sort algorithm using a queue and an indegree array. It initializes an adjacency list 'adj' and an indegree array 'indegree'. It then processes the queue, decrementing the indegree of neighbors and adding them to the queue if their indegree reaches zero. Finally, it checks if the count of processed courses equals numCourses.

The test result shows the input: numCourses = 2, prerequisites = [[1,0]]. The output is true, which matches the expected result.

LeetCode - The WorldProgress - LeetCodeCourse ScheduleSort Array by Increasing Missing Number - Le10th Largest ElementMerge Sorted ArraySort Colors - LeetCode

leetcode.com/problems/course-schedule/

Problem ListSubmit

DescriptionEditorialSolutionsSubmissions

207. Course Schedule

Solved

MediumTopicsCompaniesHint

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you must take course b_i first if you want to take course a_i.

For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

Example 1:

Input: numCourses = 2, prerequisites = [[1,0]]

Output: true

Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0. So it is possible.

Example 2:

Input: numCourses = 2, prerequisites = [[1,0],[0,1]]

Output: false

Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0, and to take course 0 you should also have finished course 1. So it is impossible.

Constraints:

1 ≤ numCourses ≤ 1000

0 ≤ prerequisites.length ≤ 1000

prerequisites[i].length == 2

0 ≤ a_i, b_i ≤ numCourses - 1

a_i ≠ b_i

18.1K350293 Online

CodeAccepted

```
43
44
45
46
47
48
49
50
51
43 indegree[neighbor]--;
44
45 if (indegree[neighbor] == 0)
46     queue[rear++] = neighbor;
47 }
48
49
50
51 return count == numCourses;
```

Saved

TestcaseTest Result

AcceptedRuntime: 0 ms

Case 1Case 2

Input

numCourses = 2

prerequisites = [[1,0],[0,1]]

Output

false

Expected

rohan_s_m_14

Access all features with our Premium subscription!

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Notebook

Progress

Points

Try New Features

Orders

My Playgrounds

Settings

Appearance

Sign Out

5 | Page

Lab Program 2:

Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>

int main() {

    int n, i, j, mobile, mobile_pos, temp;

    printf("Enter n: ");

    scanf("%d", &n);

    int a[n], dir[n];

    for(i = 0; i < n; i++) {

        a[i] = i + 1;

        dir[i] = -1;

    }

    while(1) {

        for(i = 0; i < n; i++)

            printf("%d ", a[i]);

        printf("\n");

        mobile = 0;

        mobile_pos = -1;

        for(i = 0; i < n; i++) {

            if(dir[i] == -1 && i > 0 && a[i] > a[i - 1] && a[i] > mobile) {

                mobile = a[i];

                mobile_pos = i;

            }

            if(dir[i] == 1 && i < n - 1 && a[i] > a[i + 1] && a[i] > mobile) {

                mobile = a[i];

                mobile_pos = i;

            }

        }

    }
```

```

if(mobile == 0)
    break;
if(dir[mobile_pos] == -1) {
    temp = a[mobile_pos];
    a[mobile_pos] = a[mobile_pos - 1];
    a[mobile_pos - 1] = temp;
    temp = dir[mobile_pos];
    dir[mobile_pos] = dir[mobile_pos - 1];
    dir[mobile_pos - 1] = temp;
    mobile_pos--;
} else {
    temp = a[mobile_pos];
    a[mobile_pos] = a[mobile_pos + 1];
    a[mobile_pos + 1] = temp;
    temp = dir[mobile_pos];
    dir[mobile_pos] = dir[mobile_pos + 1];
    dir[mobile_pos + 1] = temp;
    mobile_pos++;
}
for(i = 0; i < n; i++) {
    if(a[i] > mobile)
        dir[i] = -dir[i];
    }
}
return 0;
}

```

Output:

The screenshot displays the CodeBlocks IDE interface. The main editor window shows a C program named 'Topological-sorting.c'. The code implements a topological sort algorithm using a directed graph. It includes a main function that reads the number of nodes (n) and directed edges (m), then processes the edges to determine a topological order. The program uses arrays to store the graph structure and a queue to manage nodes with zero in-degree.

```
1 #include <stdio.h>
2 int main() {
3     int n, i, j, mobile, mobile_pos, temp;
4     printf("Enter n: ");
5     scanf("%d", &n);
6     int a[n], dir[n];
7     for(i = 0; i < n; i++) {
8         a[i] = i + 1;
9         dir[i] = -1;
10    }
11    while(1) {
12        for(i = 0; i < n; i++)
13            printf("%d ", a[i]);
14        printf("\n");
15        mobile = 0;
16        mobile_pos = -1;
17        for(i = 0; i < n; i++) {
18            if(dir[i] == -1 && i > 0 && a[i] > a[i-1]) {
19                mobile = a[i];
20                mobile_pos = i;
21            }
22            if(dir[i] == 1 && i < n - 1 && a[i] > a[i+1]) {
23                mobile = a[i];
24                mobile_pos = i;
25            }
26        }
27        if(mobile == 0)
28            break;
29        if(dir[mobile_pos] == -1) {
30            temp = a[mobile_pos];
31            a[mobile_pos] = a[mobile_pos - 1];
32            a[mobile_pos - 1] = temp;
33            temp = dir[mobile_pos];
34        }
35    }
```

The output window shows the execution results for n=3. The output displays the sequence of nodes in the topological order: 1 2 3, 1 3 2, 3 1 2, 3 2 1, 2 3 1, and 2 1 3. The process returned 0 (0x0) and the execution time was 11.928 s. The logs window shows the build process completed successfully with 0 errors and 0 warnings.

Lab program 3:

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

void merge(int a[], int l, int m, int r) {
    int i=l,j=m+1,k=0,temp[100000];
    while(i<=m && j<=r) {
        if(a[i]<a[j]) temp[k++]=a[i++];
        else temp[k++]=a[j++];
    }
    while(i<=m) temp[k++]=a[i++];
    while(j<=r) temp[k++]=a[j++];
    for(i=l,k=0;i<=r;i++,k++) a[i]=temp[k];
}

void mergesort(int a[], int l, int r) {
    if(l<r) {
        int m=(l+r)/2;
        mergesort(a,l,m);
        mergesort(a,m+1,r);
        merge(a,l,m,r);
    }
}

int main() {
    int n,i;
    clock_t start,end;
    double time_taken;
```

```

printf("Enter number of elements: ");

scanf("%d",&n);

int a[n];

srand(time(0));

for(i=0;i<n;i++)

    a[i]=rand()%10000;

start=clock();

mergesort(a,0,n-1);

end=clock();

time_taken=((double)(end-start))/CLOCKS_PER_SEC;

printf("Sorted elements:\n");

for(i=0;i<n;i++)

    printf("%d ",a[i]);

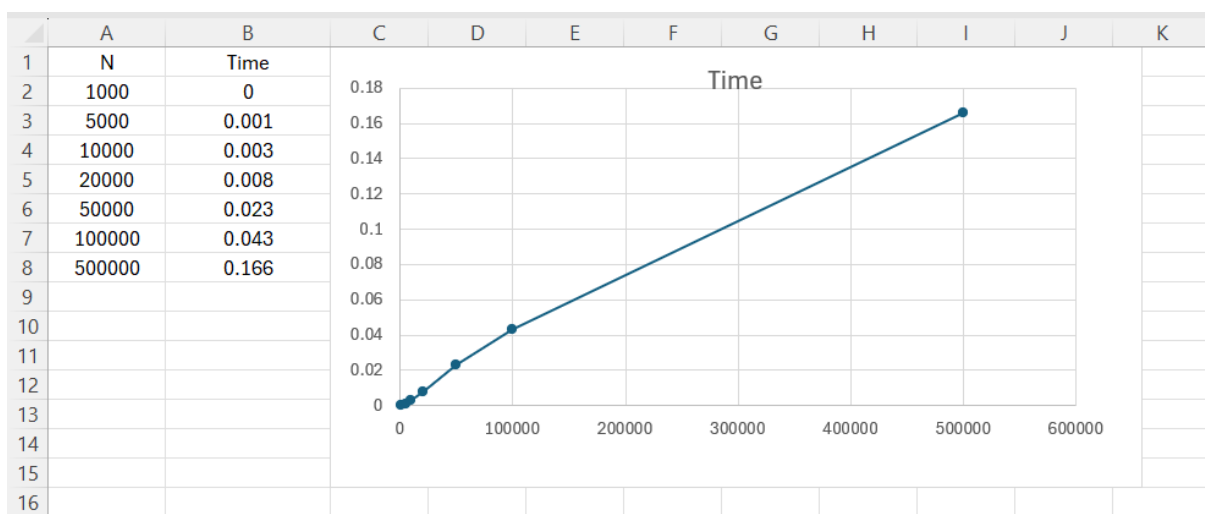
printf("\nTime taken = %f seconds\n",time_taken);

return 0;

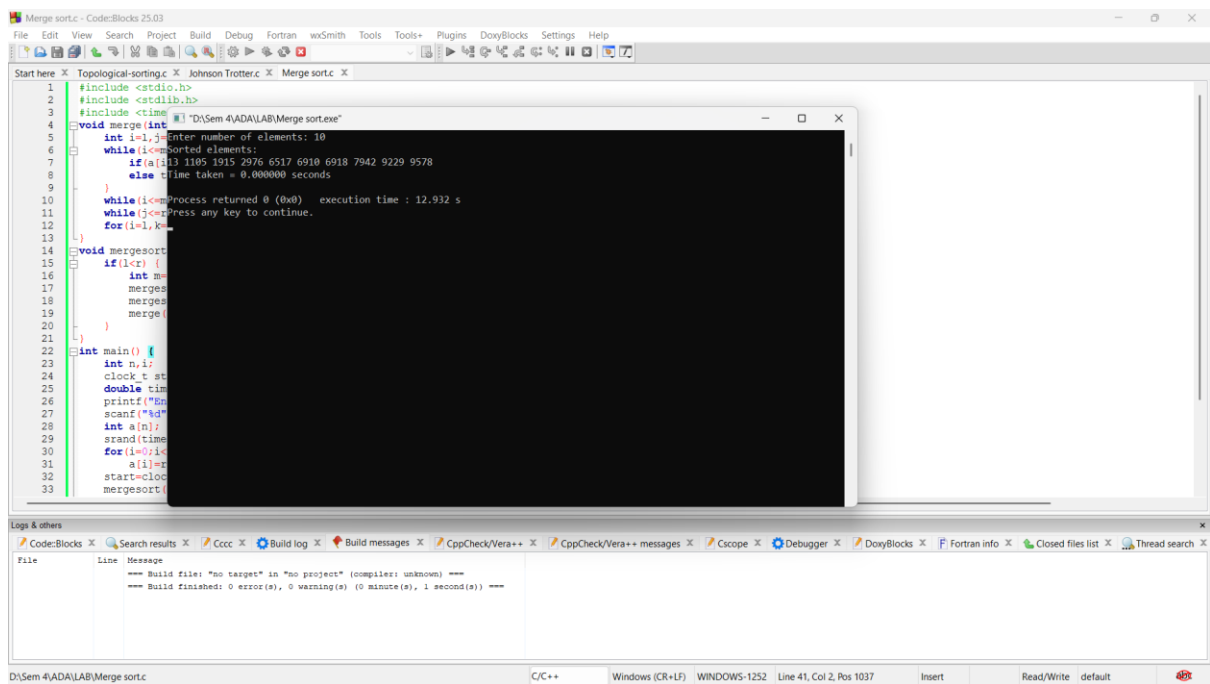
}

```

Graph:



Output:



```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <time.h>
4 void merge(int a[], int l, int m, int r) {
5     int i=l, j=m+1, k=l;
6     while(i<=m & j<=r) {
7         if(a[i]<a[j]) a[k++] = a[i++];
8         else a[k++] = a[j++];
9     }
10    while(i<=m) a[k++] = a[i++];
11    while(j<=r) a[k++] = a[j++];
12 }
13
14 void mergesort(int a[], int l, int r) {
15     if(l<r) {
16         int m = (l+r)/2;
17         mergesort(a, l, m);
18         mergesort(a, m+1, r);
19         merge(a, l, m, r);
20     }
21 }
22
23 int main() {
24     int n,i;
25     clock_t start;
26     double time;
27     printf("Enter number of elements: ");
28     scanf("%d", &n);
29     srand(time);
30     for(i=0;i<n;i++)
31         a[i]=rand()%10000;
32     start=clock();
33     mergesort(a, 0, n-1);
34     time=(double)(clock()-start)/CLOCKS_PER_SEC;
35     printf("Sorted elements: ");
36     for(i=0;i<n;i++)
37         printf("%d ", a[i]);
38     printf("\n");
39     printf("Time taken = %.6f seconds\n", time);
40     printf("Process returned 0 (0x0)   execution time : 12.932 s\n");
41     printf("Press any key to continue.\n");
42     getch();
43 }
```

Code::Blocks - Merge sort.c

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here x Topological-sorting.c x Johnson Trotter.c x Merge sort.c x

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <time.h>
4 void merge(int a[], int l, int m, int r) {
5 int i=l, j=m+1, k=l;
6 while(i<=m & j<=r) {
7 if(a[i]<a[j]) a[k++] = a[i++];
8 else a[k++] = a[j++];
9 }
10 while(i<=m) a[k++] = a[i++];
11 while(j<=r) a[k++] = a[j++];
12 }
13
14 void mergesort(int a[], int l, int r) {
15 if(l<r) {
16 int m = (l+r)/2;
17 mergesort(a, l, m);
18 mergesort(a, m+1, r);
19 merge(a, l, m, r);
20 }
21 }
22
23 int main() {
24 int n,i;
25 clock_t start;
26 double time;
27 printf("Enter number of elements: ");
28 scanf("%d", &n);
29 srand(time);
30 for(i=0;i<n;i++)
31 a[i]=rand()%10000;
32 start=clock();
33 mergesort(a, 0, n-1);
34 time=(double)(clock()-start)/CLOCKS_PER_SEC;
35 printf("Sorted elements: ");
36 for(i=0;i<n;i++)
37 printf("%d ", a[i]);
38 printf("\n");
39 printf("Time taken = %.6f seconds\n", time);
40 printf("Process returned 0 (0x0) execution time : 12.932 s\n");
41 printf("Press any key to continue.\n");
42 getch();
43 }

Logs & others

Code::Blocks x Search results x C++ x Build log x Build messages x CppCheck/Ver++ x CppCheck/Ver++ messages x Cscope x Debugger x DoxyBlocks x Fortran info x Closed files list x Thread search x

File	Line	Message
		==== Build file: "no target" in "no project" (compiler: unknown) ====
		==== Build finished: 0 error(s), 0 warning(s) (0 minute(s), 1 second(s)) ====

D:\Sem 4\ADA\LAB\Merge sort.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 41, Col 2, Pos 1037 Insert Read/Write default

Lab program 4:

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

int partition(int a[], int low, int high) {

    int pivot=a[low],i=low+1,j=high,temp;

    while(1) {

        while(i<=high && a[i]<=pivot) i++;

        while(a[j]>pivot) j--;

        if(i<j) {

            temp=a[i];

            a[i]=a[j];

            a[j]=temp;

        } else

            break;

    }

    temp=a[low];

    a[low]=a[j];

    a[j]=temp;

    return j;

}

void quicksort(int a[], int low, int high) {

    if(low<high) {

        int p=partition(a,low,high);

        quicksort(a,low,p-1);
```

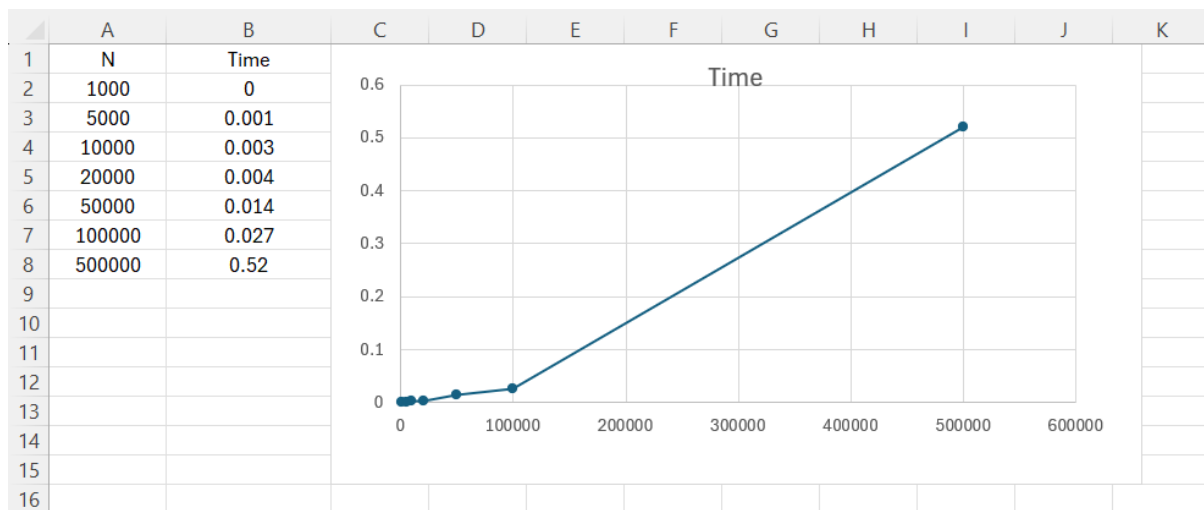
```

        quicksort(a,p+1,high);
    }
}

int main() {
    int n,i;
    clock_t start,end;
    double time_taken;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int a[n];
    srand(time(0));
    for(i=0;i<n;i++)
        a[i]=rand()%10000;
    start=clock();
    quicksort(a,0,n-1);
    end=clock();
    time_taken=((double)(end-start))/CLOCKS_PER_SEC;
    printf("Sorted elements:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);
    printf("\nTime taken = %f seconds\n",time_taken);
    return 0;
}

```


Graph:



Output:

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <time.h>
4  int partition(int a[], int low, int high)
5  {
6      int pivot=a[low],i=low+1,j=high,temp;
7      while(i<j) {
8          while(i<=high && a[i]<=pivot) i++;
9          while(a[j]>pivot) j--;
10         if(i<j) {
11             temp=a[i];
12             a[i]=a[j];
13             a[j]=temp;
14         } else break;
15     }
16     temp=a[low];
17     a[low]=a[j];
18     a[j]=temp;
19     return j;
20 }
21 void quicksort(int a[], int low, int high)
22 {
23     if(low<high) {
24         int p=partition(a,low,high);
25         quicksort(a,low,p-1);
26         quicksort(a,p+1,high);
27     }
28 }
29 int main() {
30     int n,i;
31     clock_t start,end;
32     double time_taken;
33     printf("Enter number of elements: ");
34     scanf("%d",&n);
35 }

```

Enter number of elements: 10
Sorted elements:
1230 2548 3866 4829 4859 5287 6319 7269 7349 8500
Time taken = 0.000000 seconds
Process returned 0 (0x0) execution time : 2.936 s
Press any key to continue.

Build File: "no target" in "no project" (compiler: unknown) ---
Build finished: 0 error(s), 0 warning(s) (0 minute(s), 1 second(s)) ---

Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

void heapify(int a[], int n, int i) {

    int largest=i,l=2*i+1,r=2*i+2,temp;

    if(l<n && a[l]>a[largest]) largest=l;

    if(r<n && a[r]>a[largest]) largest=r;

    if(largest!=i) {

        temp=a[i];

        a[i]=a[largest];

        a[largest]=temp;

        heapify(a,n,largest);

    }

}

void heapsort(int a[], int n) {

    int i,temp;

    for(i=n/2-1;i>=0;i--)

        heapify(a,n,i);

    for(i=n-1;i>0;i--) {

        temp=a[0];

        a[0]=a[i];

        a[i]=temp;

        heapify(a,i,0);

    }

}
```

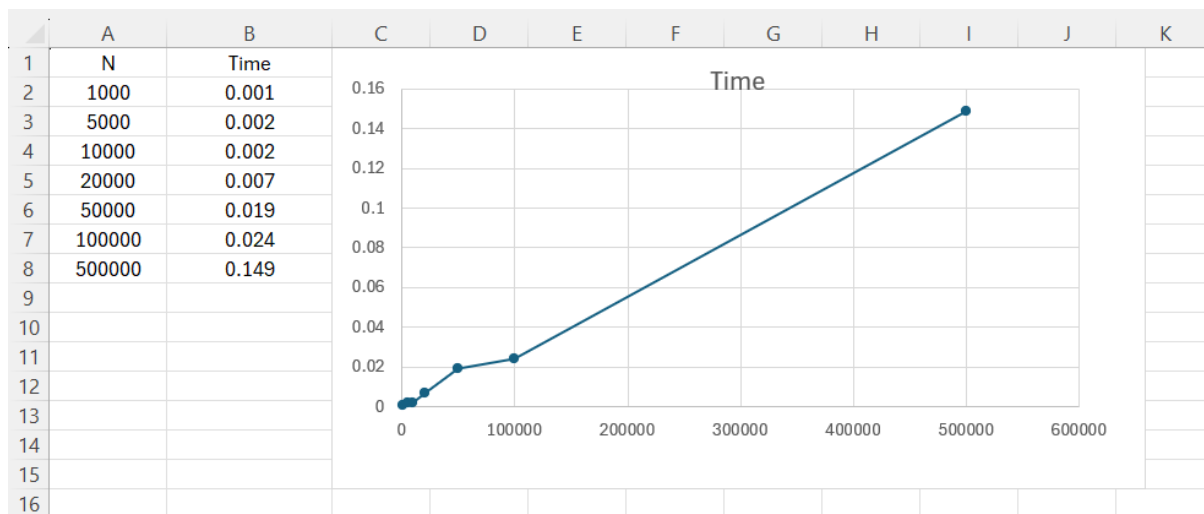
```

}

int main() {
    int n,i;
    clock_t start,end;
    double time_taken;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int a[n];
    srand(time(0));
    for(i=0;i<n;i++)
        a[i]=rand()%10000;
    start=clock();
    heapsort(a,n);
    end=clock();
    time_taken=((double)(end-start))/CLOCKS_PER_SEC;
    printf("Sorted elements:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);
    printf("\nTime taken = %f seconds\n",time_taken);
    return 0;
}

```

Graph:



Output:

The screenshot shows the Code::Blocks IDE with a C++ project named 'Heap sort'. The source code implements the heap sort algorithm. The output window shows the following results:

```

Enter number of elements: 10
Sorted elements:
405 989 1902 3222 3549 4472 4654 7329 8707 9078
Time taken = 0.000000 seconds
Process returned 0 (0x0)   execution time : 3.984 s
Press any key to continue.
  
```

The code includes headers for `<stdio.h>`, `<stdlib.h>`, and `<time.h>`. It defines functions `heapify` and `heapsort`, and a `main` function that reads the number of elements, performs the sort, and prints the sorted array and time taken.

Lab program 6:

Implement 0/1 Knapsack problem using dynamic programming.

```
#include <stdio.h>

int main() {
    int n, W, i, j;

    printf("Enter number of items: ");
    scanf("%d", &n);

    int wt[n+1], val[n+1];
    printf("Enter weights:\n");
    for(i=1;i<=n;i++)
        scanf("%d",&wt[i]);

    printf("Enter profits:\n");
    for(i=1;i<=n;i++)
        scanf("%d",&val[i]);

    printf("Enter capacity: ");
    scanf("%d",&W);

    int dp[n+1][W+1];
    for(i=0;i<=n;i++) {
        for(j=0;j<=W;j++) {
            if(i==0 || j==0)
                dp[i][j]=0;
            else if(wt[i]<=j) {
                int include=val[i]+dp[i-1][j-wt[i]];
                int exclude=dp[i-1][j];
                dp[i][j]=(include>exclude)?include:exclude;
            } else
                dp[i][j]=dp[i-1][j];
        }
    }
}
```

```

    }

}

printf("\nDP Table:\n");

for(i=0;i<=n;i++) {

    for(j=0;j<=W;j++)

        printf("%4d",dp[i][j]);

    printf("\n");

}

printf("\nMaximum Profit = %d\n",dp[n][W]);

return 0;

}

```

Output:

```

1 #include <stdio.h>
2 int main() {
3     int n, w, i, j;
4     printf("Enter number of items: ");
5     scanf("%d", &n);
6     int wt[n+1], val[n+1];
7     printf("Enter weights: ");
8     for(i=1;i<=n;i++)
9         scanf("%d", &wt[i]);
10    printf("Enter profits: ");
11    for(i=1;i<=n;i++)
12        scanf("%d", &val[i]);
13    printf("Enter capacity: ");
14    scanf("%d", &w);
15    int dp[n+1][w+1];
16    for(i=0;i<=n;i++)
17        for(j=0;j<=w;j++)
18            dp[i][j] = 0;
19    printf("Maximum Profit = ");
20    printf("%d", dp[n][w]);
21    return 0;
22 }

```

Output:

```

Enter number of items: 4
Enter weights: 2 3 4 5
Enter profits: 3 4 5 6
Enter capacity: 7
Maximum Profit = 9

```

Leet Code exercise on Knapsack problem using dynamic programming

879. Profitable Schemes

Problem:

There is a group of n members, and a list of various crimes they could commit. The i^{th} crime generates a $\text{profit}[i]$ and requires $\text{group}[i]$ members to participate in it. If a member participates in one crime, that member can't participate in another crime.

Let's call a profitable scheme any subset of these crimes that generates at least minProfit profit, and the total number of members participating in that subset of crimes is at most n .

Return the number of schemes that can be chosen. Since the answer may be very large, return it modulo $10^9 + 7$.

Code:

```
#define MOD 1000000007

int profitableSchemes(int n, int minProfit, int* group, int groupSize, int* profit, int profitSize)
{
    int dp[101][101] = {0};
    dp[0][0] = 1;
    for (int k = 0; k < groupSize; k++) {
        int g = group[k];
        int p = profit[k];
        for (int members = n; members >= g; members--) {
            for (int prof = minProfit; prof >= 0; prof--) {
                int newProfit = prof + p;
                if (newProfit > minProfit)
                    newProfit = minProfit;
                dp[members][newProfit] =
                    (dp[members][newProfit] +
                     dp[members - g][prof]) % MOD;
            }
        }
    }
}
```

```

long long ans = 0;

for (int members = 0; members <= n; members++) {

    ans = (ans + dp[members][minProfit]) % MOD;

}

return (int)ans;
}

```

Output:

The screenshot shows the LeetCode interface for the 'Profitable Schemes' problem (879). The problem description is on the left, and the code editor is on the right. The code is in C++ and implements a dynamic programming solution. The test results show 'Accepted' with a runtime of 0 ms.

```

1  Define MOD 1000000007
2  int profitableSchemes(int n, int minProfit, int* group, int groupSize,
3
4      int dp[101][101] = {0};
5
6      dp[0][0] = 1;
7
8      for (int k = 0; k < groupSize; k++) {
9          for (int i = 0; i < n; i++) {
10             for (int j = 0; j < minProfit; j++) {
11                 int g = group[k];
12                 int p = profit[k];
13
14                 for (int members = i; members >= g; members--) {
15                     for (int prof = minProfit; prof >= j + p; prof--) {
16                         int newProfit = prof + p;
17                         dp[members][newProfit] = (dp[members][newProfit] + dp[members - g][j]) % MOD;
18                     }
19                 }
20             }
21         }
22     }
23
24     long long ans = 0;
25     for (int members = 0; members <= n; members++) {
26         ans = (ans + dp[members][minProfit]) % MOD;
27     }
28
29     return (int)ans;
30 }

```

The screenshot shows the LeetCode interface for the 'Profitable Schemes' problem (879). The problem description is on the left, and the code editor is on the right. The code is in C++ and implements a dynamic programming solution. The test results show 'Accepted' with a runtime of 1 ms.

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```


Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include <stdio.h>

#define INF 999

int main() {
    int n, i, j, k;

    printf("Enter number of vertices: ");

    scanf("%d", &n);

    int a[n][n];

    printf("Enter cost matrix:\n");

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(k=0;k<n;k++)
        for(i=0;i<n;i++)
            for(j=0;j<n;j++)
                if(a[i][k]+a[k][j]<a[i][j])
                    a[i][j]=a[i][k]+a[k][j];

    printf("\nAll Pair Shortest Path Matrix:\n");

    for(i=0;i<n;i++) {
        for(j=0;j<n;j++)
            printf("%4d",a[i][j]);

        printf("\n");
    }

    return 0;
}
```

Output:

The screenshot shows a C++ IDE with a code editor on the left and a console window on the right. The code implements the Floyd-Warshall algorithm for finding all-pair shortest paths. The console output shows the input (4 vertices, a cost matrix) and the resulting shortest path matrix.

```
1 #include <stdio.h>
2 #define INF 999
3
4 int main() {
5     int n, i, j, k;
6     printf("Enter number of vertices: ");
7     scanf("%d", &n);
8     int a[n][n];
9     printf("Enter cost matrix:\n");
10    for(i=0; i<n; i++)
11        for(j=0; j<n; j++)
12            scanf("%d", &a[i][j]);
13    for(k=0; k<n; k++)
14        for(i=0; i<n; i++)
15            for(j=0; j<n; j++)
16                if(a[i][k]+a[k][j]<a[i][j])
17                    a[i][j]=a[i][k]+a[k][j];
18    printf("\nAll Pair Shortest Path Matrix:\n");
19    for(i=0; i<n; i++) {
20        for(j=0; j<n; j++)
21            printf("%d\t", a[i][j]);
22        printf("\n");
23    }
24    return 0;
25 }
```

Console Output:

```
Enter number of vertices: 4
Enter cost matrix:
0 5 999 10
999 0 3 999
999 999 0 1
999 999 999 0

All Pair Shortest Path Matrix:
0 5 8 9
999 0 3 4
999 999 0 1
999 999 999 0

Process returned 0 (0x0)   execution time : 418.096 s
Press any key to continue.
```

Leet Code exercise on Floyd's algorithm

1334. Find the City With the Smallest Number of Neighbors at a Threshold Distance

Problem:

There are n cities numbered from 0 to $n-1$. Given the array `edges` where `edges[i] = [fromi, toi, weighti]` represents a bidirectional and weighted edge between cities `fromi` and `toi`, and given the integer `distanceThreshold`.

Return the city with the smallest number of cities that are reachable through some path and whose distance is at most `distanceThreshold`. If there are multiple such cities, return the city with the greatest number.

Notice that the distance of a path connecting cities i and j is equal to the sum of the edges' weights along that path.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#define INF 1000000000
int findTheCity(int n, int** edges, int edgesSize, int* edgesColSize, int distanceThreshold)
{
    int dist[105][105];
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i == j)
                dist[i][j] = 0;
            else
                dist[i][j] = INF;
        }
    }
    // Fill edges
    for (int i = 0; i < edgesSize; i++) {
        int u = edges[i][0];
        int v = edges[i][1];
        int w = edges[i][2];
```

```

    dist[u][v] = w;
    dist[v][u] = w;
}
// Floyd-Warshall
for (int via = 0; via < n; via++) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {

            if (dist[i][via] + dist[via][j] < dist[i][j]) {
                dist[i][j] = dist[i][via] + dist[via][j];
            }
        }
    }
}

int minReachable = INF;
int answer = -1;
for (int i = 0; i < n; i++) {
    int count = 0;
    for (int j = 0; j < n; j++) {
        if (dist[i][j] <= distanceThreshold) {
            count++;
        }
    }
    // Exclude itself
    count--;
    // If tie, choose greater city number
    if (count <= minReachable) {
        minReachable = count;
        answer = i;
    }
}

```

```

    }

    return answer;
}

```

Output:

The screenshot shows the LeetCode interface for problem 1334. The problem description states: "There are n cities numbered from 0 to $n-1$. Given the array `edges` where `edges[i] = [fromi, toi, weighti]` represents a bidirectional and weighted edge between cities `fromi` and `toi`, and given the integer `distanceThreshold`. Return the city with the smallest number of cities that are reachable through some path and whose distance is **at most** `distanceThreshold`. If there are multiple such cities, return the city with the greatest number. Notice that the distance of a path connecting cities i and j is equal to the sum of the edges' weights along that path."

Example 1:

```

graph LR
    0 ---|3| 1
    1 ---|1| 2
    1 ---|4| 3
    2 ---|1| 3

```

The code editor shows a C++ solution:

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define INF 100000000
5
6 int findTheCity(int n, int** edges, int edgesSize, int edgesColSize, int distanceThreshold) {
7
8     int dist[105][105];
9
10    // Initialize distance matrix
11    for (int i = 0; i < n; i++) {
12        for (int j = 0; j < n; j++) {

```

The test case shows `n = 4` and `edges = [[0,1,3],[1,2,1],[1,3,4],[2,3,1]]`. The result is "Accepted" with a runtime of 0 ms.

The screenshot shows the same LeetCode problem page, but with a different C++ solution in the code editor:

```

59 // If tie, choose greater city number
60 if (count <= minReachable) {
61     minReachable = count;
62     answer = i;
63 }
64 }

```

The test case shows `n = 5` and `edges = [[0,1,2],[0,4,8],[1,2,3],[1,4,2],[2,3,1],[3,4,1]]`. The `distanceThreshold` is `2`. The output is `0`, which matches the expected result.

Lab program 8:

(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>

#define INF 999

int main() {
    int n, i, j, min, u, v, total = 0;
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    int cost[n][n], visited[n];
    printf("Enter cost matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&cost[i][j]);
    for(i=0;i<n;i++)
        visited[i]=0;
    visited[0]=1;
    printf("\nEdges in MST:\n");
    for(int e=0;e<n-1;e++) {
        min=INF;
        for(i=0;i<n;i++) {
            if(visited[i]) {
                for(j=0;j<n;j++) {
                    if(!visited[j] && cost[i][j]<min) {
                        min=cost[i][j];
                        u=i;
                        v=j;
                    }
                }
            }
        }
    }
}
```

```

printf("%d -> %d : %d\n",u,v,min);

total+=min;

visited[v]=1;

}

printf("Minimum Cost = %d\n",total);

return 0;

}

```

Output:

```

D:\Sem 4\ADA\LAB\Prim's algorithm.exe
Enter number of vertices: 5
Enter cost matrix:
2 999 6 999
999 3 8 5
999 3 999 999 7
8 999 999 9
999 5 7 9 999
Edges in MST:
0 -> 1 : 2
0 -> 2 : 3
1 -> 4 : 5
0 -> 3 : 6
Minimum Cost = 16
Process returned 0 (0x0)   execution time : 145.626 s
Press any key to continue.

```

(b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>

#define INF 999

int parent[20];

int find(int i) {
    while(parent[i])
        i=parent[i];
    return i;
}

int uni(int i,int j) {
    if(i!=j) {
        parent[j]=i;
        return 1;
    }
    return 0;
}

int main() {
    int n,i,j,min,a,b,u,v,total=0,edges=0;
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    int cost[20][20];
    printf("Enter cost matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&cost[i][j]);
    printf("\nEdges in MST:\n");
    while(edges<n-1) {
        min=INF;
        for(i=0;i<n;i++) {
            for(j=0;j<n;j++) {
```



```

        if(cost[i][j]<min) {
            min=cost[i][j];
            a=u=i;
            b=v=j;
        }
        u=find(u);
        v=find(v);
        if(uni(u,v)) {
            printf("%d -> %d : %d\n",a,b,min);
            total+=min;
            edges++;
        }
        cost[a][b]=cost[b][a]=INF;
    }
    printf("Minimum Cost = %d\n",total);
    return 0;
}

```

Output:

The screenshot shows the Code::Blocks IDE with the following components:

- Main Window:** Displays the C++ code for Kruskal's algorithm. The code includes headers, function definitions for `find`, `uni`, and `main`, and a `main` function that reads input, calculates the MST, and prints the result.
- Terminal Window:** Shows the program's output. It prompts for the number of vertices (5) and the cost matrix. The output displays the edges in the MST and the final minimum cost (16).
- Status Bar:** Shows the file path `D:\Sem 4\ADA\LAB\Kruskal's algorithm.c`, the compiler `C/C++`, and the line/col information `Line 48, Col 2, Pos 1092`.

Lab program 9:

Implement Fractional Knapsack using Greedy technique.

```
#include <stdio.h>

struct Item {
    int weight, profit;
    float ratio;
};

int main() {
    int n, i, j, capacity;
    float maxProfit = 0.0;
    printf("Enter number of items: ");
    scanf("%d", &n);
    struct Item item[n], temp;
    printf("Enter weights and profits:\n");
    for(i=0;i<n;i++) {
        scanf("%d%d",&item[i].weight,&item[i].profit);
        item[i].ratio=(float)item[i].profit/item[i].weight;
    }
    printf("Enter knapsack capacity: ");
    scanf("%d",&capacity);
    for(i=0;i<n-1;i++) {
        for(j=0;j<n-i-1;j++) {
            if(item[j].ratio<item[j+1].ratio) {
                temp=item[j];
                item[j]=item[j+1];
                item[j+1]=temp;
            }
        }
    }
    for(i=0;i<n;i++) {
```

```

    if(capacity>=item[i].weight) {
        capacity-=item[i].weight;
        maxProfit+=item[i].profit;
    } else {
        maxProfit+=item[i].ratio*capacity;
        break;
    }
}

printf("Maximum Profit = %.2f\n",maxProfit);

return 0;
}

```

Output:

The screenshot shows the CodeBlocks IDE with a C++ project named 'Fractional Knapsack'. The source code is visible in the editor, and the console window shows the program's execution. The program prompts the user to enter the number of items (3), weights (10, 20, 30), and profits (60, 100, 120). It then calculates the maximum profit as 240.00.

```

Enter number of items: 3
Enter weights and profits:
Enter number: 10 20 30
Enter profits: 60 100 120
Enter knapsack capacity: 50
Maximum Profit = 240.00
Process returned 0 (0x0)   execution time : 27.170 s
Press any key to continue.

```

Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>

#define INF 999

int main() {
    int n, i, j, u, source;

    printf("Enter number of vertices: ");
    scanf("%d",&n);

    int cost[n][n], dist[n], visited[n];

    printf("Enter cost matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&cost[i][j]);

    printf("Enter source vertex: ");
    scanf("%d",&source);

    for(i=0;i<n;i++) {
        dist[i]=cost[source][i];
        visited[i]=0;
    }

    dist[source]=0;
    visited[source]=1;

    for(i=1;i<n;i++) {
        int min=INF;

        for(j=0;j<n;j++) {
            if(!visited[j] && dist[j]<min) {
                min=dist[j];
```

```

        u=j;
    }
}

visited[u]=1;

for(j=0;j<n;j++) {

    if(!visited[j] && dist[u]+cost[u][j]<dist[j])

        dist[j]=dist[u]+cost[u][j];

}

}

printf("\nShortest distances from vertex %d:\n",source);

for(i=0;i<n;i++)

    printf("%d -> %d = %d\n",source,i,dist[i]);

return 0;

}

```

Output:

The screenshot shows a C++ IDE with the following code in 'Dijkstra's Algorithm.c':

```

1 #include <stdio.h>
2 #define INF 999
3 int main() {
4     int n, i, j, u, source;
5     printf("Enter number of vertices: ");
6     scanf("%d", &n);
7     int cost[n][n], dist[n], visited[n];
8     printf("Enter cost matrix:\n");
9     for(i=0; i<n; i++)
10         for(j=0; j<n; j++)
11             scanf("%d", &cost[i][j]);
12     printf("Enter source vertex: ");
13     scanf("%d", &source);
14     for(i=0; i<n; i++) {
15         dist[i]=cost[source][i];
16         visited[i]=0;
17     }
18     dist[source]=0;
19     visited[source]=1;
20     for(i=1; i<n; i++) {
21         int min=INF;
22         for(j=0; j<n; j++) {
23             if(!visited[j] && dist[j]<min) {
24                 min=dist[j];
25                 u=j;
26             }
27         }
28         visited[u]=1;
29         for(j=0; j<n; j++) {
30             if(!visited[j] && dist[u]+cost[u][j]<dist[j])
31                 dist[j]=dist[u]+cost[u][j];
32         }
33     }
34 }

```

The output window shows the following results:

```

Enter number of vertices: 5
Enter cost matrix:
0 10 999 30 100
10 0 50 999 999
999 50 0 20 10
100 999 10 60 0
Enter source vertex: 0
Shortest distances from vertex 0:
0 -> 0 = 0
0 -> 1 = 10
0 -> 2 = 50
0 -> 3 = 30
0 -> 4 = 60
Process returned 0 (0x0)   execution time : 40.849 s
Press any key to continue.

```

Lab program 11:

Implement “N-Queens Problem” using Backtracking

```
#include <stdio.h>

#include <stdlib.h>

int x[20], n;

int place(int k, int i) {
    int j;
    for(j=1;j<k;j++)
        if(x[j]==i || abs(x[j]-i)==abs(j-k))
            return 0;
    return 1;
}

void printSolution() {
    int i,j;
    printf("\nSolution:\n");
    for(i=1;i<=n;i++) {
        for(j=1;j<=n;j++) {
            if(x[i]==j)
                printf(" Q ");
            else
                printf(" . ");
        }
        printf("\n");
    }
}

void nQueens(int k) {
    int i;
    for(i=1;i<=n;i++) {
```

```

        if(place(k,i)) {

            x[k]=i;

            if(k==n)

                printSolution();

            else

                nQueens(k+1);

        }

    }

}

int main() {

    printf("Enter value of N: ");

    scanf("%d",&n);

    nQueens(1);

    return 0;

}.

```

Output:

The screenshot shows the Code::Blocks IDE with the following components:

- Editor:** Displays the source code for `NQueens.c`. The code includes headers, defines an array `x` of size `n`, and implements functions `place`, `printSolution`, and `nQueens` to solve the N-Queens problem.
- Output Window:** Shows the execution results. It prompts for the value of `N` (4) and displays the solutions for the 4-Queens problem:


```

Solution:
. Q . .
. . . Q
Q . . .
. . Q .
      
```
- Log & Messages Window:** Shows the build process. It indicates that the build was successful with 0 errors and 0 warnings, and that the execution time was 16.002 seconds.

Leetcode exercises :

75. Sort colors

Problem:

Given an array `nums` with `n` objects colored red, white, or blue, sort them in place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Code:

```
#include <stdio.h>

void sortColors(int* nums, int numsSize) {
    int low = 0, mid = 0, high = numsSize - 1;
    while (mid <= high) {
        if (nums[mid] == 0) {
            int temp = nums[low];
            nums[low] = nums[mid];
            nums[mid] = temp;
            low++;
            mid++;
        } else if (nums[mid] == 1) {
            mid++;
        } else {
            int temp = nums[mid];
            nums[mid] = nums[high];
            nums[high] = temp;
            high--;
        }
    }
}
```


Output:

The screenshot shows the LeetCode interface for the 'Sort Colors' problem (75). The problem description states: 'Given an array `nums` with `n` objects colored red, white, or blue, sort them *in-place* so that objects of the same color are adjacent, with the colors in the order red, white, and blue. We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively. You must solve this problem without using the library's sort function.'

Example 1:
Input: `nums = [2,0,2,1,1,0]`
Output: `[0,0,1,1,2,2]`

Example 2:
Input: `nums = [2,0,1]`
Output: `[0,1,2]`

Constraints:

- `n == nums.length`
- `1 <= n <= 300`
- `nums[i]` is either 0, 1, or 2.

Follow up: Could you come up with a one-pass algorithm using only constant extra space?

The 'Code' tab shows a C++ solution using a two-pass algorithm (Dutch National Flag):

```
1 #include <stdio.h>
2
3 void sortColors(int* nums, int numsSize) {
4     int low = 0, mid = 0, high = numsSize - 1;
5
6     while (mid <= high) {
7         if (nums[mid] == 0) {
8             int temp = nums[low];
9             nums[low] = nums[mid];
10            nums[mid] = temp;
11            low++;
12            mid++;
13        } else if (nums[mid] == 1) {
14            mid++;
15        } else {
16            int temp = nums[mid];
17            nums[mid] = nums[high];
18            nums[high] = temp;
19            high--;
20        }
21    }
22 }
```

The 'Testcase' tab shows the test results for Case 1 and Case 2, both of which are 'Accepted'.

The screenshot shows the LeetCode interface for the 'Sort Colors' problem (75). The problem description is the same as in the first screenshot.

Example 1:
Input: `nums = [2,0,2,1,1,0]`
Output: `[0,0,1,1,2,2]`

Example 2:
Input: `nums = [2,0,1]`
Output: `[0,1,2]`

Constraints:

- `n == nums.length`
- `1 <= n <= 300`
- `nums[i]` is either 0, 1, or 2.

The 'Code' tab shows a Java solution using a two-pass algorithm (Dutch National Flag):

```
11
12
13     } else if (nums[mid] == 1) {
14         mid++;
15     } else {
16         int temp = nums[mid];
17         nums[mid] = nums[high];
18         nums[high] = temp;
19         high--;
20     }
21 }
22 }
```

The 'Testcase' tab shows the test results for Case 1 and Case 2, both of which are 'Accepted'.

88. Merged sorted array

Problem:

You are given two integer arrays `nums1` and `nums2`, sorted in non-decreasing order, and two integers `m` and `n`, representing the number of elements in `nums1` and `nums2` respectively.

Merge `nums1` and `nums2` into a single array sorted in non-decreasing order.

The final sorted array should not be returned by the function, but instead be *stored inside the array* `nums1`. To accommodate this, `nums1` has a length of `m + n`, where the first `m` elements denote the elements that should be merged, and the last `n` elements are set to 0 and should be ignored. `nums2` has a length of `n`

Code:

```
void merge(int* nums1, int nums1Size, int m, int* nums2, int nums2Size, int n) {
    int i = m - 1;
    int j = n - 1;
    int k = m + n - 1;
    while (i >= 0 && j >= 0) {
        if (nums1[i] > nums2[j]) {
            nums1[k] = nums1[i];
            i--;
        } else {
            nums1[k] = nums2[j];
            j--;
        }
        k--;
    }
    while (j >= 0) {
        nums1[k] = nums2[j];
        j--;
        k--;
    }
}
```

Output:

88. Merge Sorted Array

You are given two integer arrays `nums1` and `nums2`, sorted in **non-decreasing order**, and two integers `m` and `n`, representing the number of elements in `nums1` and `nums2` respectively.

Merge `nums1` and `nums2` into a single array sorted in **non-decreasing order**.

The final sorted array should not be returned by the function, but instead be stored **inside the array `nums1`**. To accommodate this, `nums1` has a length of `m + n`, where the first `m` elements denote the elements that should be merged, and the last `n` elements are set to `0` and should be ignored. `nums2` has a length of `n`.

Example 1:

Input: `nums1 = [1,2,3,0,0,0]`, `m = 3`, `nums2 = [2,5,6]`, `n = 3`
Output: `[1,2,2,3,5,6]`
Explanation: The arrays we are merging are `[1,2,3]` and `[2,5,6]`.
The result of the merge is `[1,2,2,3,5,6]` with the underlined elements coming from `nums1`.

Example 2:

Input: `nums1 = [1]`, `m = 1`, `nums2 = []`, `n = 0`
Output: `[1]`
Explanation: The arrays we are merging are `[1]` and `[]`.
The result of the merge is `[1]`.

Example 3:

Input: `nums1 = [0]`, `m = 0`, `nums2 = [1]`, `n = 1`
Output: `[1]`
Explanation: The arrays we are merging are `[]` and `[1]`.
Note that because `m = 0`, there are no elements in `nums1`. The `0` is only there to ensure the merge result can fit in `nums1`.

Constraints:

- `nums1.length == m + n`

```
void merge(int* nums1, int nums1Size, int m, int* nums2, int nums2Size, int n) {
    int i = m - 1;
    int j = n - 1;
    int k = m + n - 1;

    while (i >= 0 && j >= 0) {
        if (nums1[i] > nums2[j]) {
            nums1[k] = nums1[i];
            i--;
        } else {
            nums1[k] = nums2[j];
            j--;
        }
        k--;
    }
    while (j >= 0) {
        nums1[k] = nums2[j];
        j--;
        k--;
    }
}
```

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

`nums1 =`
`[1,2,3,0,0,0]`

`m =`
`3`

`nums2 =`
`[2,5,6]`

`n =`
`3`

Sort Colors - LeetCode

You are given an array `nums` containing `n` numbers sorted in ascending order. Sort the array in-place according to the following rules:

- 0s should come first
- 1s should come next
- 2s should come last

Return the sorted array.

Example 1:

Input: `nums = [0,1,0,1,0,1]`
Output: `[0,0,0,1,1,1]`

Example 2:

Input: `nums = [2,0,1,1,0]`
Output: `[0,0,1,1,2]`

Constraints:

- `0 <= nums[i] <= 2`

```
void sortColors(int* nums, int numsSize) {
    int i = 0;
    int j = 0;
    int k = 0;

    while (i < numsSize) {
        if (nums[i] == 0) {
            nums[i] = nums[k];
            nums[k] = 0;
            k++;
        } else if (nums[i] == 1) {
            nums[i] = nums[j];
            nums[j] = 1;
            j++;
        } else if (nums[i] == 2) {
            nums[i] = nums[k];
            nums[k] = 2;
            k++;
        }
        i++;
    }
}
```

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

`nums =`
`[1]`

`m =`
`1`

`nums2 =`
`[1]`

`n =`
`0`

Output

`[1]`

215. Kth largest element in an array

Problem:

Given an integer array *nums* and an integer *k*, return *the kth largest element in the array*.

Note that it is the *kth* largest element in the sorted order, not the *kth* distinct element.

Code:

```
#include <stdio.h>
#include <stdlib.h>

void swap(int* a, int* b) {    int temp = *a;
    *a = *b;
    *b = temp;
}

void heapify(int* heap, int n, int i) {
    int smallest = i;
    int left = 2*i + 1;
    int right = 2*i + 2;
    if (left < n && heap[left] < heap[smallest]) smallest = left;
    if (right < n && heap[right] < heap[smallest]) smallest = right;
    if (smallest != i) {
        swap(&heap[i], &heap[smallest]);
        heapify(heap, n, smallest);
    }
}

int findKthLargest(int* nums, int numsSize, int k) {
    int* heap = (int*)malloc(sizeof(int) * k);
    int heapSize = 0;
    for (int i = 0; i < numsSize; i++) {
        if (heapSize < k) {
            heap[heapSize++] = nums[i];
        }
        if (heapSize == k) {
            heapify(heap, k, 0);
        }
    }
    return heap[0];
}
```

```

        for (int j = k/2 - 1; j >= 0; j--) {
            heapify(heap, k, j);
        }
    }

    if (nums[i] > heap[0]) {
        heap[0] = nums[i];
        heapify(heap, k, 0);
    }
}

int result = heap[0];
free(heap);
return result;
}

```

Output:

The screenshot shows a web browser displaying the LeetCode problem '215. Kth Largest Element in an Array'. The problem description states: 'Given an integer array `nums` and an integer `k`, return the k^{th} largest element in the array. Note that it is the k^{th} largest element in the sorted order, not the k^{th} distinct element. Can you solve it without sorting?'. Examples provided are: Example 1: Input: `nums = [3,2,1,5,6,4]`, `k = 2`, Output: `5`; Example 2: Input: `nums = [3,2,3,1,2,4,5,6]`, `k = 4`, Output: `4`. Constraints: $1 \leq k \leq \text{nums.length} \leq 10^5$ and $-10^6 \leq \text{nums}[i] \leq 10^6$. The solution code is written in C++ and uses a min-heap. The code is as follows:

```

17 if (right < n && heap[right] < heap[smallest]) smallest = right;
18
19 if (smallest != i) {
20     swap(heap[i], heap[smallest]);
21     heapify(heap, n, smallest);
22 }
23
24
25 int findKthLargest(int* nums, int numsSize, int k) {
26     int* heap = (int*)malloc(sizeof(int) * k);
27     int heapSize = k;
28
29     for (int i = 0; i < numsSize; i++) {
30         if (heapSize < k) {
31             heap[heapSize++] = nums[i];
32             if (heapSize == k) {
33                 for (int j = k/2 - 1; j >= 0; j--) {

```

The test case shows the input `nums = [3,2,1,5,6,4]` and `k = 2`, with the expected output `5`. The solution is marked as 'Accepted' with a runtime of 0 ms.

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Progress - LeetCode

Kth Largest Element in an Array

Merge Sorted Array - LeetCode

Sort Colors - LeetCode

leetcode.com/problems/kth-largest-element-in-an-array/

Problem List

Submissions

Status	Language	Runtime	Memory	Notes
Accepted	C	24 ms	15.8 MB	
Accepted	C	20 ms	15.8 MB	

Code

```
31 heap[heapSize++] = nums[i];
32 if (heapSize == k) {
33     for (int j = k/2 - 1; j > 0; j--)
34         heapify(heap, k, j);
35 }
36 }
37 }
38 } else if (nums[i] > heap[0]) {
39     heap[0] = nums[i];
40     heapify(heap, k, 0);
41 }
42 }
```

Accepted

Runtime: 0 ms

Case 1 Case 2

Input

nums = [3,2,3,1,2,4,5,6]

k = 4

Output

4

rohan_s_m_14

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268. Missing number

Problem:

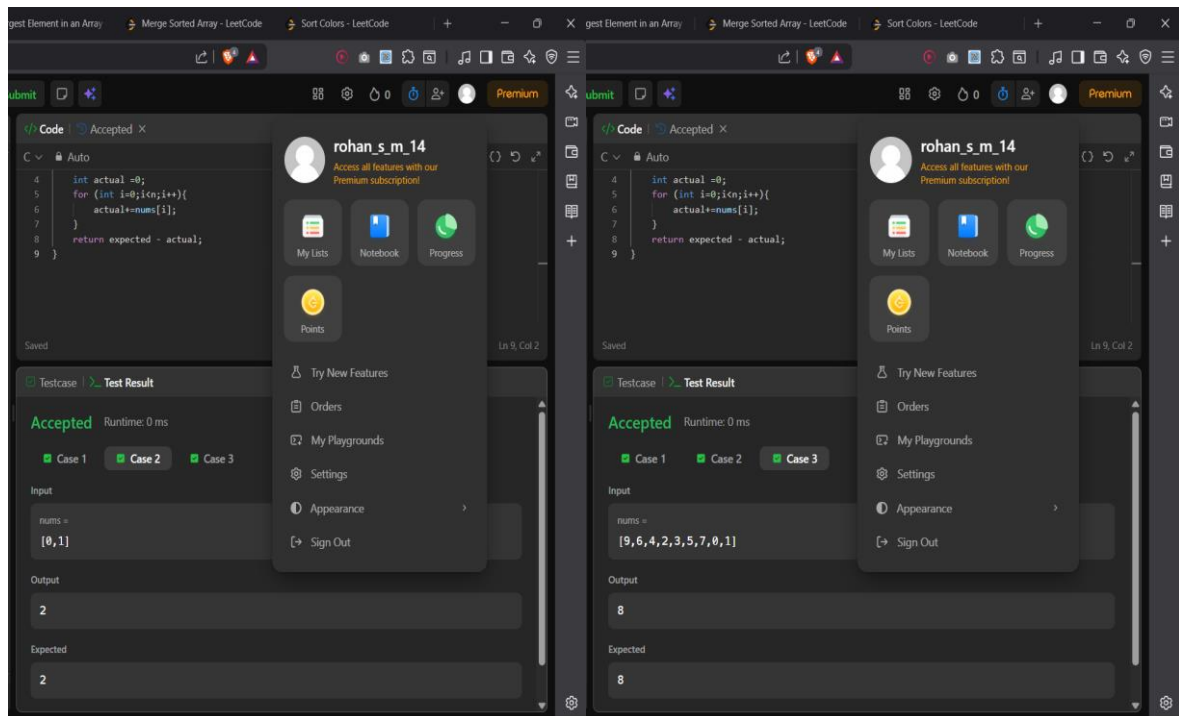
Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return the only number in the range that is missing from the array.

Code:

```
int missingNumber(int* nums, int numsSize) {  
    int n=numsSize;  
    int expected = (n * (n+1))/2;  
    int actual =0;  
    for (int i=0;i<n;i++){  
        actual+=nums[i];  
    }  
    return expected - actual;  
}
```

Output:

The screenshot displays the LeetCode website interface for problem 268, "Missing Number". The left sidebar shows the problem description, which states: "Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return the only number in the range that is missing from the array." It includes three examples: Example 1 (Input: `nums = [3,0,1]`, Output: `2`), Example 2 (Input: `nums = [0,1]`, Output: `2`), and Example 3 (Input: `nums = [9,6,4,2,3,5,7,0,1]`, Output: `8`). The main content area shows the C code solution for the problem, which is the same as the code provided in the previous block. The code is marked as "Accepted" and has a runtime of 0 ms. The right sidebar shows the user's profile, "rohan_s_m_14", and a list of navigation links including "My Lists", "Notebook", "Progress", "Points", "Try New Features", "Orders", "My Playgrounds", "Settings", "Appearance", and "Sign Out".



1636. Sort Array by Increasing Frequency

Problem:

Given an array of integers `nums`, sort the array in increasing order based on the frequency of the values. If multiple values have the same frequency, sort them in decreasing order.

Return the sorted array.

Code:

```
#include <stdio.h>
#include <stdlib.h>

int* frequencySort(int* nums, int numsSize, int* returnSize) {
    int freq[201] = {0};
    int minIdx = 201, maxIdx = -1, maxFreq = 0;
    for (int i = 0; i < numsSize; i++) {
        int idx = nums[i] + 100;
        freq[idx]++;
        if (freq[idx] > maxFreq) maxFreq = freq[idx];
        if (idx < minIdx) minIdx = idx;
        if (idx > maxIdx) maxIdx = idx;
    }
    int* buckets[201] = {0};
    int bucketSizes[201] = {0};
    for (int i = minIdx; i <= maxIdx; i++) {
        if (freq[i] == 0) continue;
        int f = freq[i];
        if (!buckets[f]) buckets[f] = (int*)malloc(sizeof(int) * numsSize);
        for (int j = 0; j < f; j++) {
            buckets[f][bucketSizes[f]++] = i - 100;
        }
    }
}
```

```

int* result = (int*)malloc(sizeof(int) * numsSize);

int idx = 0;

for (int f = 0; f <= maxFreq; f++) {
    if (bucketSizes[f]) {
        for (int j = bucketSizes[f] - 1; j >= 0; j--) {
            result[idx++] = buckets[f][j];
        }
        free(buckets[f]);
    }
}

*returnSize = numsSize;

return result;
}

```

Output:

The screenshot displays the LeetCode interface for the problem "1636. Sort Array by Increasing Frequency". The problem description states: "Given an array of integers `nums`, sort the array in increasing order based on the frequency of the values. If multiple values have the same frequency, sort them in decreasing order." Examples provided include:

- Example 1: Input: `nums = [1,1,2,2,2,3]`, Output: `[3,1,1,2,2,2]`. Explanation: '3' has a frequency of 1, '1' has a frequency of 2, and '2' has a frequency of 3.
- Example 2: Input: `nums = [2,3,1,3,2,2]`, Output: `[1,3,3,2,2,2]`. Explanation: '2' and '3' both have a frequency of 2, so they are sorted in decreasing order.
- Example 3: Input: `nums = [-1,1,-6,4,5,-6,1,4,1]`, Output: `[5,-1,4,4,-6,-6,1,1,1]`.

 Constraints: `1 <= nums.length <= 100` and `-100 <= nums[i] <= 100`. The solution code is in C, using a bucket sort approach. The code is shown in the 'Code' tab, and the 'Test Result' tab shows 'Accepted' with a runtime of 0 ms. The sidebar on the right shows the user's profile 'rohan_s_m_14' and various navigation options.

